Arturo de Giorgi

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Personal Data

Nationality and Birth:	Italy | 1996
PhD:	10/2024
Currently:	Postdoctoral Researcher at Durham University (IPPP), UK
Previously:	PhD Student at IFT UAM/CSIC Madrid (2021 - 2024), ES
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Research Interests

Theoretical Particle Physics — BSM Phenomenology.

My research focuses on the phenomenology of physics beyond the Standard Model, with emphasis on extra-dimensional scenarios and their cosmological and astrophysical signatures. A central theme is the Kaluza-Klein (KK) gravitational sector: its spin-2, vector, and scalar towers, the sum rules and symmetry structures governing their interactions, and their interplay with dark matter, axions, and neutrinos across collider, astrophysical, and cosmological scales.

Keywords: *BSM, Extra Dimensions, Kaluza-Klein Phenomenology, Massive Spin-2, Dark Matter, Axions/ALPs, Neutrinos, Gravitational Waves.*

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Education & Academic Stays

Education

Ph.D. in Physics, Instituto de Física Teórica (IFT UAM/CSIC), Madrid 09/2021–09/2024
Final Grade: *Cum Laude*, International Mention; Advisor: Prof. Dr. Luca Merlo
Thesis: *The Phenomenological Landscape of Leptophilic ALPs*

M.Sc. in Physics, LMU Munich 10/2018–12/2020
Final Grade: 1.02/1.00; Advisors: Dr. G. Raffelt and S. Vogl
Thesis: *Dark Matter Production in Warped Extra-Dimensions*

B.Sc. in Physics, University of Padova 09/2015–07/2018
Final Grade: 109/110; Advisor: Prof. Dr. P. Marchetti
Thesis: *Non-Abelian Anyons and Quantum Computation*

Academic Stays (2+ Months)

5. Visitor via QUPIP Grant at **QUP-KEK**, Tsukuba, Japan, 04–06/2025
   in the group of *Dr. V. Takhistov*
4. Visitor at **Ruprecht-Karls-Universität Heidelberg**, 10–12/2023
   in the group of *Prof. Dr. J. Jaeckel*
3. Visitor at **Harvard University**, Boston, 06–08/2023
   in the group of *Prof. Dr. C. Argüelles-Delgado*
2. Research Internship at **ETH**, Zurich, 05–08/2021
   in the group of *Prof. Dr. M.K. Marinkovic*
1. Student Research Assistant at **Max Planck Institut für Physik**, Munich, 01–11/2020
   in the group of *Dr. Habil. G. Raffelt*

Grants, Prizes & Honors

Prizes and Awards

09/2026 | **Award for Excellence in Research**
Awarded by Durham University Physics Department.
"For outstanding research excellence through the development of a highly successful programme in physics beyond the Standard Model, spanning axions, dark matter, neutrino physics, gravitational waves and novel astrophysical signatures."

09/2025 | **2025 Prize in Theoretical Physics "Giuliano Preparata"**
Awarded by the Italian Physical Society (SIF).
"For significant contributions to theoretical and phenomenological aspects of physics beyond the Standard Model, in particular on new low-mass particles and extra dimensions."

Grants

04/2025 | QUP Internship Program (QUPIP)
International Center for Studies of the Universe and Particles (WPI-QUP), KEK, Tsukuba, Japan.
See: QUPIP.

09/2021 | Doctoral Grant — European Union's Horizon 2020, **Marie Skłodowska-Curie** grant

agreement No 860881-HIDDeN, as Early Stage Researcher (ESR).
See: hiddeneu.eu.

Minor Awards

- 1st Prize in Poster Competition at Invisible24 School 07/2024
- 1st Prize in Poster Competition at Invisible23 School 09/2023
- Best Poster Presentation in Poster Competition at Invisible22 School 07/2022
- Scholarship for high school students "Premio Gelati" (€1,000) 03/2016

Community & Leadership

Collaborations

1. COST Action CA21106 COSMIC WISPers:
   - **Co-Leader Working Group 1 (Theory)** 04/2025 – 10/2026
   - **Grants Evaluation Committee** 09/2024 – 10/2026
   - **Young Researcher Council Representative** 07/2024 – 05/2025
   - Monthly **Colloquium Organiser** 01/2023 – 07/2024

Editorial Experience

- **Journal Referee** for *Europhysics Letters (EPL)* 10/2026 – Today
- **Journal Referee** for the *Journal of High Energy Physics (JHEP)* 06/2026 – Today
- **Guest Editor** for *Physics Letters B (PLB)* 03/2026 – Today
- **Journal Referee** for the *European Physical Journal C (EPJC)* 03/2025 – Today
- **Editor** for Proceedings of the 2nd Training School of COST Action COSMIC WISPers (COSMICWISPers2024) 06/2024

Organization of International Events

- **International Organizing Committee**, CA21106 4th General Meeting Corfu Summer Institute, Corfu, Greece 06/2026
- **Junior Local Organizing Committee**, Extended Workshop NuTs 2022 Institute for Theoretical Physics (IFT), Madrid, Spain 05/2022

Talks at Conferences & Workshops

Plenary Sessions

9. **14th KIAS Workshop & 6th AEI Workshop for BSM**, KIAS, South Korea.
   Stochastic Tsunamis: Diffuse Scalar Background from Black Hole Formation 09/2026
8. **ALPs Across Scales**, MITP - Mainz Institute for Theoretical Physics, Germany
   Hunting Multiple Axions 06/2026
7. **5th Asian-European-Institutes Workshop for BSM**, Durham University, UK.
   New BSM transient signals from black hole formation 09/2025
6. **Evolving Beyond the SM**, CERN, Geneva, Switzerland.
   New BSM transient signals from black hole formation 07/2025
5. **QUP Workshop**, KEK, Tsukuba, Japan.
   Star shearing season: transient signals from black hole formation 04/2025

4. **Invisibles24 Workshop**, University of Bologna, Italy.
   Extra-Dimensional Maxion't 07/2024

3. **Invisibles23 Workshop**, University of Göttingen, Germany.
   Down to the Seesaw Line via the JALZ ALP-HNL Portal 09/2023

2. **57th Rencontres de Moriond**, La Thuille, Italy. Session: Electroweak Interactions & Unified Theories.
   The Low Seesaw Scale Solution for M_W and $(g-2)_\mu$ 03/2023

1. **Invisibles22 Workshop**, IJCLab, Orsay, France.
   BSM flavoured correlations 06/2022

Parallel Sessions

5. **Pascos 2025**, Durham University, UK.
   Star shearing season: transient signals from black hole formation 07/2025

4. **21st Lomonosov Conference**, Moscow State University, Russia.
   The low-scale seesaw solution to the M_W and $(g-2)_\mu$ anomalies 08/2023

3. **Planck 2023**, Warsaw, Poland.
   BSM and 2HDM: update, prospects and a bridge to ALPs 05/2023

2. **DESY Theory Workshop**, DESY, Hamburg, Germany.
   Spin-2 mediated Dark Matter in Warped Extra-Dimensions 09/2021

1. **2021 Phenomenology Symposium**, University of Pittsburgh, USA.
   Spin-2 mediated Dark Matter in Warped Extra-Dimensions 05/2021

Invited Seminars

16. *The Hunt for Multiple Axions.* IFT, Madrid, July, 2026
15. *The Hunt for Multiple Axions.* DESY, Germany, May, 2026
14. *Hunting Multiple Axions.* Tsung-Dao Lee Institute, China, Mar., 2026
13. *Hunting for Multiple Axions.* University of Basel, Switzerland, Feb., 2026
12. *Transient Signals from Black Hole formation.* KEK Theory Center, Japan, June, 2025
11. *Extra-dimensional axion patterns.* KEK Theory Center, Japan, Apr., 2025
10. *The Extradimensional Portal to Dark Matter.* Institute of Cosmos Sciences of the University of Barcelona (ICCUB), May, 2024
9. *The Minimal Massive Majoron.* Institute for Theoretical Physics, University Heidelberg, Dec., 2023
8. *KK-Gravitons Sum Rules and Phenomenology.* University of Zurich, Nov., 2023
7. *Flavour and Higgs physics in Z2-symmetric 2HD models near the decoupling limit.* LHC Higgs Working Group - WG2 WG3 joint meeting on CP violation in extended Higgs sector, Sept., 2023
6. *Down to the Seesaw Line via the JALZ ALP-HNL Portal.* Harvard University, June, 2023
5. *An old and a new opportunity for ALP physics.* Karlsruher Institut für Technologie, June, 2023
4. *An Opportunity in the ALP-HNLs Sector.* COST Cosmic WISPers - WG1 Meeting, May, 2023
3. *A Piece Beyond HNLs.* Albert-Ludwigs-Universität Freiburg, May, 2023
2. *BSM Higgs Flavoured Correlation.* LHC Higgs Working Group - Common WG2 and WG3 – CP violation and Higgs Sector, June, 2022

1. *Spin-2 mediated Dark Matter in Warped Extra-Dimensions.* Albert-Ludwigs-Universität Freiburg, July, 2021

Teaching

- **Tutor** at the 2026 STFC HEP Summer School (UK national PhD training school in HEP) — 09/2026
- **Tutor** at the 2025 STFC HEP Summer School (UK national PhD training school in HEP) — 09/2025
- **Tutor** for Level 1 course, Durham University Foundations of Physics 1 *(40 hours)* — 10/2025–04/2026
- **Tutor** for Level 1 course, Durham University Foundations of Physics 1 *(40 hours)* — 10/2024–04/2025
- **Tutor** for Master course, Ludwig-Maximilians-Universität München Classical and Quantum Simulations of Physical Systems, held by *Prof. Dr. M.K. Marinkovic* — 10–12/2020
- **Tutor** for Master course, Ludwig-Maximilians-Universität München Introduction to Lattice Gauge Theories, held by *Prof. Dr. M.K. Marinkovic* — 07–09/2020
- **Tutor** for Master course, Ludwig-Maximilians-Universität München Quantum Mechanics II, held by *Prof. Dr. V. Mukhanov* — 10/2019–03/2020

Publications

I have authored 25 papers, published (or in progress to be) in prestigious peer-reviewed journals, 4 community white papers, and 3 proceedings, with several further works in preparation. My research has been recognised with the 2025 SIF Prize in Theoretical Physics and Durham University's Physics Department Excellence Award.

Publications

25. **A. de Giorgi** and S. Vogl, *Vector Dark Matter from $d \geq 2$ Extra Dimensions*, arXiv:2610.05851
24. **A. de Giorgi** and D. Pasari, *The Dark Dimension and Majorana Neutrinos*, arXiv:2609.10668
23. **A. de Giorgi** and J. Jaeckel, *Stochastic Tsunamis: Diffuse Scalar Background from Black Hole Formation*, arXiv:2607.12013
22. **A. de Giorgi**, M. Marcoli, and F. Silvetti, *LHC Constraints on Resonant Kaluza-Klein Gravitons*, arXiv:2607.12012
21. **A. de Giorgi**, J. Ingoldby, V. V. Khoze, and J. Turner, *Thermal Metastable Strings in One-Scale Models and Gravitational Waves*, arXiv:2606.02689
20. **A. de Giorgi**, D. Naredo-Tuero, and X. Ponce Díaz, *The Majoron Cosmological Window: Dark Matter and Thermal Leptogenesis*, arXiv:2605.18944
19. **A. de Giorgi**, Y. G. del Castillo, and J. Jaeckel, *Scalar tsunamis from black hole formation*, JCAP **09** (2026) 127, [arXiv:2602.06112]
18. **A. de Giorgi**, J. Jaeckel, S. Monath, and V. Takhistov, *Multiple axions in laboratory experiments*, JHEP **09** (2026) 177, [arXiv:2512.16837]
17. **A. de Giorgi**, I. M. Soler, and S. Sevillano Muñoz, *Neutrino Constraints on Scalar-Tensor Gravity*, arXiv:2512.13798

16. **A. de Giorgi**, D. Pasari, and J. Turner, *Do neutrinos dream in 5D? Towards a comprehensive extra-dimensional neutrino phenomenology*, JHEP **05** (2026) 152, [arXiv:2512.02101]

15. **A. de Giorgi** and M. Ramos, *Extra-dimensional axion patterns*, Phys. Rev. D **111** (2025), no. 7 075006, [arXiv:2412.00179]

14. **A. de Giorgi** and J. Jaeckel, *Star shearing season: transient signals in wave-like dark matter experiments from black hole formation*, Eur. Phys. J. C **85** (2025), no. 2 155, [arXiv:2408.10296]

13. M. B. Marcos, **A. de Giorgi**, L. Merlo, and J.-L. Tastet, *ALPs and HNLs at LHC and muon colliders: Uncovering new couplings and signals*, SciPost Phys. **18** (2025) 084, [arXiv:2407.14970]

12. **A. de Giorgi**, M. F. Zamoro, and L. Merlo, *Visible GeV ALP from TeV Vector-Like Leptons*, Fortsch. Phys. **73** (2025), no. 3 2400165, [arXiv:2402.14059]

11. **A. de Giorgi** and S. Vogl, *Gravity-matter sum rules in models with a single extra-dimension*, JHEP **05** (2024) 315, [arXiv:2311.01507]

10. **A. de Giorgi**, L. Merlo, X. Ponce Díaz, and S. Rigolin, *The minimal massive Majoron Seesaw Model*, JHEP **03** (2024) 094, [arXiv:2312.13417]

9. **A. de Giorgi**, F. Koutroulis, L. Merlo, and S. Pokorski, *Flavour and Higgs physics in Z2-symmetric 2HD models near the decoupling limit*, Nucl. Phys. B **994** (2023) 116323, [arXiv:2304.10560]

8. **A. de Giorgi**, L. Merlo, and J.-L. Tastet, *Probing HNL-ALP couplings at colliders*, Fortsch. Phys. **71** (2023), no. 4-5 2300027, [arXiv:2212.11290]

7. **A. de Giorgi** and G. Piazza, *A lesson from $R_{\tau\tau}^{K^{(*)}}$ and $R_{\nu\nu}^{K^{(*)}}$ at Belle II*, Fortsch. Phys. (2024) [arXiv:2211.05595]

6. **A. de Giorgi**, L. Merlo, and S. Pokorski, *The Low-Scale Seesaw Solution to the M_W and $(g-2)_\mu$ Anomalies*, Fortsch. Phys. **71** (2023), no. 4-5 2300020, [arXiv:2211.03797]

5. J. Bonilla, **A. de Giorgi**, B. Gavela, L. Merlo, and M. Ramos, *The cost of an ALP solution to the neutral B-anomalies*, JHEP **02** (2023) 138, [arXiv:2209.11247]

4. **A. de Giorgi** and S. Vogl, *Warm dark matter from a gravitational freeze-in in extra dimensions*, JHEP **04** (2023) 032, [arXiv:2208.03153]

3. J. Alonso-Gonzalez, **A. de Giorgi**, L. Merlo, and S. Pokorski, *Searching for BSM physics in Yukawa couplings and flavour symmetries*, JHEP **05** (2022) 041, [arXiv:2109.07490]

2. **A. de Giorgi** and S. Vogl, *Dark matter interacting via a massive spin-2 mediator in warped extra-dimensions*, JHEP **11** (2021) 036, [arXiv:2105.06794]

1. **A. de Giorgi** and S. Vogl, *Unitarity in KK-graviton production: A case study in warped extra-dimensions*, JHEP **04** (2021) 143, [arXiv:2012.09672]

White Papers and Community Inputs

4. A. Arza *et. al.*, *The COSMIC WISPers White Paper: The physics case for Weakly Interacting Slim Particles*, arXiv:2603.03433 - **Corresponding author for WG1: WISPs Model Building**

3. C. Albertus *et. al.*, *WISPedia – the WISPs Encyclopedia*, arXiv:2602.09089 - **Corresponding author**

2. D. Aybas *et. al.*, *Exploring the Dark Universe: A European Strategy for Axions and other WISPs Discovery*, arXiv:2503.20432

1. J.-H. Arling *et. al.*, *Early Career Researcher Input to the European Strategy for Particle Physics Update: White Paper*, arXiv:2503.19862

Proceedings

3. **A. de Giorgi**, M. Fuentes Zamoro, and L. Merlo, *GeV ALP from TeV Vector-like leptons*, in *Contribution to 2nd Training School and General Meeting of the COST Action COSMIC WISPers (CA21106) (COSMICWISPers2024) 10-14 June 2024 and 3-6 September 2024 Ljubljana (Slovenia) and Istanbul (Turkey)*, 11, 2024. arXiv:2411.18246

2. **A. de Giorgi**, L. Merlo, and S. Pokorski, *The Low-Scale Seesaw Solution to the M_W and $(g-2)_\mu$ Anomalies*, Moscow Univ. Phys. Bull. **79** (2024), no. Suppl 1 465–469

1. **A. de Giorgi**, L. Merlo, and S. Pokorski, *Low Seesaw Scale Solution for M_W and $(g-2)_\mu$*, in *57th Rencontres de Moriond on Electroweak Interactions and Unified Theories*, 4, 2023. arXiv:2304.08438

Outreach

6. **Podcast** on Particle Physics hosted by Dr. Shounak Bhattacharya 06/2025
Currently viewed 970+ times. Youtube Link

5. **Public Lecture** "*Why neutrinos matter*" for the International Masterclass in Particle Physics 2025, Durham University, hosting over 110 students from six local high schools. Read more. 03/2025

4. **Podcast** on Particle Physics hosted by Dr. Shounak Bhattacharya 01/2025
Currently viewed 520+ times. Youtube Link

3. **Speaker**, Jornadas de la Física, Parque de Atracciones de Madrid, ∼1200 students 03/2024

2. **Speaker**, Jornadas de la Física, Parque de Atracciones de Madrid, ∼1200 students 03/2023

1. **A. de Giorgi**, *Physics, following the breadcrumbs of our intuition*, February, 2022

Non-Scientific Activities

May 2023 – Today	Member, **Rotaract** Madrid Serrano. Community service organisation focused on international service projects.
May 2023 – Today	Mentor, **Lead the Future**. Non-profit STEM mentorship organisation based on the spirit of 'Give Back'.
Sep 2022 – May 2023	Member, IFT **Equity, Diversity and Inclusion Committee**. EDI@IFT
2014 – 2021	Founder and CAO, **Artupia** (art-tech startup).
2012 – 2015	Member and Events Coordinator, **Europeanclubeuropeo**. Club enhancing awareness of EU values of peace and cooperation among nations.

Languages and Computer Skills

Languages

Native: Italian | **Fluent:** English, Spanish | **Basic:** German

Computer Skills

Basic Knowledge:	HTML, CSS
Intermediate Knowledge:	C++, FeynCalc, FeynRules, xAct, Flavio
Advanced Knowledge:	Mathematica, LaTeX, Python

Arturo de Giorgi October 12, 2026